Fasal Kranti: Transforming Farming with Data

Real Impact, Cost, and Feasibility Analysis

The Agricultural Challenge

Indian agriculture faces persistent challenges that limit productivity and profitability for farmers across the country.

Unpredictable Weather

Erratic rainfall patterns and extreme weather events make traditional farming calendars unreliable

Pest Attacks

Unforeseen pest infestations can destroy entire crops without early warning systems

Resource Overuse

Excessive pesticide and water usage due to lack of precise measurement and guidance

Data Gap

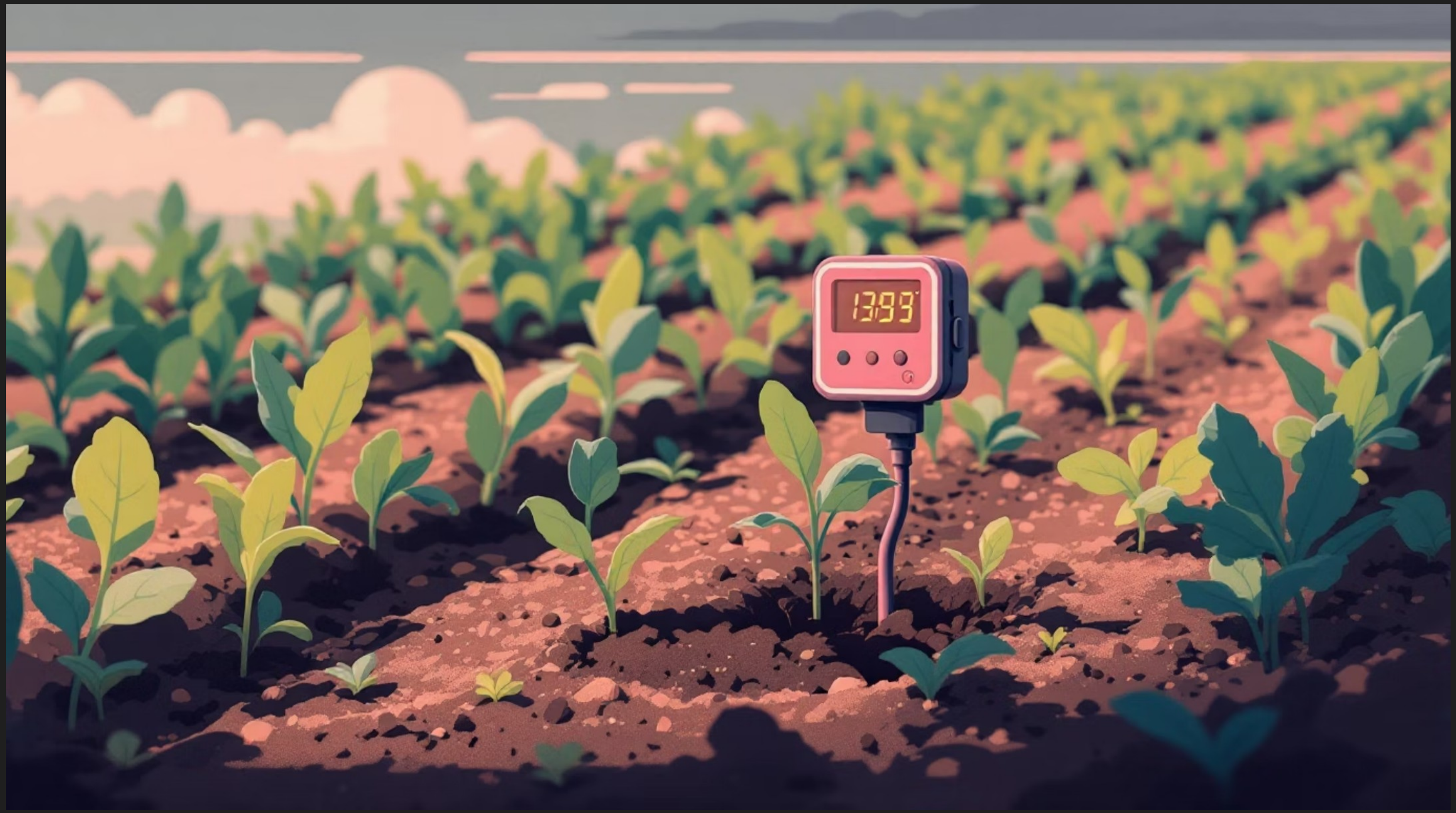
Absence of real-time data prevents informed decision-making at critical growth stages

What is Fasal Kranti

IoT-Based Sensor System

Smart sensors deployed across farmland continuously monitor critical environmental parameters and provide AI-powered recommendations through a mobile application.

- Soil moisture levels
- Air humidity and temperature
- Leaf wetness detection
- Weather forecasting integration



How the System Works

01

Deployment

Sensors installed strategically across 3–5 acres per device, covering the entire farm area efficiently

02

Data Collection

Real-time measurements transmitted to cloud servers for processing and analysis

03

AI Analysis

Machine learning algorithms process data and generate crop-specific recommendations

04

Farmer Action

Mobile app delivers actionable insights on irrigation, pesticide application, and harvest timing

Target Market & Crop Focus

Grapes

High-value export crop requiring precise water and nutrient management

Pomegranate

Premium domestic and international market with quality-sensitive pricing

Mango

Seasonal high-value production needing optimal harvest timing

Banana

Year-round cultivation with continuous monitoring requirements

Focus on horticulture crops with sufficient margin to justify technology investment. Not initially targeted at low-margin staples like rice and wheat.

Adoption & Scale Achieved

10K+

Minimum Coverage

Conservative estimate of minimum area under monitoring

60K+

Maximum Coverage

Upper estimate of potential total deployment

1000s

Farmers Served

Thousands of farmers across multiple states

Karnataka

Grapes and pomegranate belts in northern regions

Maharashtra

Diverse horticulture including mango and banana cultivation

Madhya Pradesh

Emerging pomegranate and vegetable farming areas

Chhattisgarh

Expanding horticulture adoption with government support

Cost Analysis

Investment Breakdown

The system requires upfront hardware investment with recurring subscription fees for data services and AI recommendations.

Hardware cost: ₹25,000–₹50,000 per device
Coverage: 3–5 acres per installation
Subscription: Annual fee for cloud services
Installation: Professional setup and calibration

Cost per acre: ₹5,000–₹15,000 annually

Revenue Impact Analysis

15%

Minimum Yield Increase

Lower bound of productivity improvement

40%

Maximum Yield Increase

Upper bound with optimal implementation

15%

Minimum Cost Reduction

Lower bound of input savings

30%

Maximum Cost Reduction

Upper bound in pesticides and water

Net income increase: Typically 25–40% after accounting for technology costs. Returns usually realised within first year for medium and large farms.

Case Study: Before & After Comparison

Traditional Farming

- Yield: 8 tons/acre
- Pesticides: ₹25,000/acre
- Water: 500 hours/acre
- Labour: 120 days/acre

With Fasal Kranti

- Yield: 11 tons/acre (+37.5%)
- Pesticides: ₹18,000/acre (-28%)
- Water: 400 hours/acre (-20%)
- Labour: 100 days/acre (-17%)

Net profit increase: ₹50,000–₹1,00,000 per acre annually

Feasibility & Future Outlook

Why It Succeeded

- Targeted high-value crops with clear ROI
- Designed for Indian farming conditions
- Simple mobile interface for farmers
- Data-driven recommendations

Current Limitations

- Initial cost barrier for small farmers
- Requires proper usage and training
- Less effective for low-margin crops

Future Scope

- Expansion to additional crops
- Government subsidy programmes
- Integration with supply chain (Fasal Fresh)
- Partnerships with input suppliers

> Conclusion: Precision agriculture through IoT technology demonstrates clear profitability improvements. Fasal Kranti represents a practical, scalable example of agri-tech adoption in Indian horticulture, with strong potential for expanded deployment across commercial farming segments.